The Energy Balance Project

Homeostatic Field Model and Local Actor-Motion Law

Foundation Document — Version 2.1 (Implementation and Empirical Verdict)

Status: V2.0 formal model implemented, verified, and tested. This revision records one equation correction, two modeling extensions required to make the model run, the verification results, long-horizon experiments, and an assessment of whether the proposed local movement law is a good model.

Headline result. Purely local, potential-driven rules — with no central optimizer — sustain a distributed field in its viable range indefinitely (verified to 50,000 ticks), but only inside a bounded window of renewable supply. Outside that window the field collapses, exactly as Law 6 requires. The model is a sound scientific instrument; it is not, and does not claim to be, a guarantee of survival.

Abstract

This document reports the first executable realization of the V2.0 homeostatic field model. The physics engine — a single dynamic capacity field on a lattice, governed by the seven hard laws (locality, continuity, bounded state, non-negative dissipation, bounded regeneration, no enforced homeostasis, and counterfactual separation) — was implemented in Python and validated against the eight unit tests specified in V2.0 Section 14.1, all of which pass, including the ideal lossless monotonicity proof (dB/dt <= 0). We then asked the project's central question: can the local movement law keep the field viable forever? Using a producer/consumer checkerboard world of 100 cells and one local actor per cell, the gradient-flow rule holds 94–95% of cells in the viable band with mean burden ~0.33 across 200, 5,000, and 50,000 ticks, whereas the identical world with redistribution disabled loses 50% of cells permanently. Implementation revealed that the V2.0 fixed-leakage term is insufficient for stable in-band homeostasis; we introduce a state-dependent (proportional) leakage term as a self-regulating sink. We conclude with a verdict against the model's own falsification criteria.

1. Purpose and Relationship to V2.0

V2.0 defined a formal model and an implementation specification but was explicitly a “formal model under development.” This V2.1 does not change the conceptual commitments of V2.0. It records what happened when the specification was built and executed: which equations were implemented verbatim, which required correction or extension to produce a runnable and stable simulation, and what the simulation actually does over long horizons. All V2.0 hard laws are preserved. Following the V2.0 discipline, every added quantity below solves a demonstrated modeling failure rather than merely adding realism.

2. Implemented Model (faithful to V2.0)

The world is a graph G = (V, E); the first implementation uses an n x n square lattice with a von Neumann four-neighborhood. Each cell i carries one dynamic state x_i(t) >= 0, the usable functional capacity. The no-action counterfactual update (V2.0 Sec. 2) is implemented exactly:

```
x_i^0(t+1) = clip_[0,K_i] { x_i + s_i + g_i(x_i) - d_i - lambda_i }
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The homeostatic burden functional and marginal potential (V2.0 Sec. 6) are implemented verbatim:

```
ell_i(x_i) = alpha_i [L_i - x_i]_+^2 + beta_i [x_i - U_i]_+^2
B(x)       = sum_i ell_i(x_i)

mu_i = -2 alpha_i (L_i - x_i)   if x_i < L_i
mu_i = 0                        if L_i <= x_i <= U_i
mu_i =  2 beta_i  (x_i - U_i)   if x_i > U_i
```

The local movement law (V2.0 Sec. 7) drives transfers by the generalized force and clips them to the physical constraints:

```
F_ij = mu_i - mu_j - theta_ij
q_ij = min { q_a^max,  x_i - x_i^min,  K_j - x_j,  M_ij [F_ij]_+ }
delivered = eta_ij q_ij ,   transport loss = (1 - eta_ij) q_ij + c_ij^0 >= 0
```

A deterministic actor selects the neighboring edge of greatest positive force and otherwise rests. Continuity (Law 2), bounded state (Law 3), non-negative dissipation (Law 4), and counterfactual separation (Law 7 — a no-action branch is created every tick and natural flows are never credited to actors) are enforced by the engine, not by actor preference. Oversubscribed sources are resolved by proportional scaling (V2.0 Sec. 12.1).

3. Corrections and Extensions to the Equations

Three changes were required to move from specification to a runnable, stable model. Only the first is a correction to an equation; the other two are concrete realizations of choices V2.0 left open.

3.1 Correction: state-dependent leakage (the key fix)

V2.0 lists leakage lambda_i(t) as an externally specified function but the worked examples treat it as a fixed per-tick constant. With fixed leakage the model exhibits no stable in-band attractor: an open system either bleeds out (net drain, slow collapse invisible at short horizons) or accumulates to the capacity bound K_i and hoards. Neither is homeostasis. We therefore make leakage state-dependent — larger stocks leak more — which is the minimal self-regulating sink:

```
lambda_i = lambda_i^0 + kappa_i * x_i(t)          (kappa_i >= 0)
```

This single change gives the field a stable fixed point strictly inside [L_i, U_i]: inflow is balanced against demand plus a loss that grows with capacity, so cells settle in-band instead of emptying or saturating. It is fully consistent with V2.0 (leakage was always permitted to be any function of state) and it is what converts “survives” into “stays healthy.”

3.2 Extension: a dense field of local actors

V2.0 specifies ~50 mobile actors but leaves their deployment open. To test the movement law as a pure local field rule — the “Game of Life” reading of the project — we place one stationary actor on every cell, each applying the identical law using only its four-neighborhood. No actor sees the global state and nothing is centrally coordinated. This is the strictest possible test of the claim that local rules alone produce global viability.

3.3 Extension: the producer/consumer checkerboard test world

To exercise transport under locality (Law 1) with minimal confounding transport loss, sources and sinks are arranged as a checkerboard: producer cells (renewable inflow s > 0, no demand) alternate with consumer cells (demand d > 0, no inflow). Every consumer is surrounded by producers, so the gradient rule only ever needs one-hop transfers. This isolates the question “does the local rule route supply to need?” from the separate question of long-distance transport loss.

Not yet implemented (scope): the H-tick regeneration-aware counterfactual (V2.0 Sec. 9, Model D), Allee-threshold regenerative sources, the softmax exploration policy, and the EBU ledger overlay (Model E). The current actor is Model C (gradient proposal, one-tick acceptance).

4. Verification (V2.0 Section 14.1)

All eight required pre-experiment unit tests pass. The decisive one is the monotonicity proof of V2.0 Section 8: under ideal lossless gradient flow the burden never increases, and in our run it was driven to exactly zero.

#	Required property (V2.0 Sec. 14.1)	Result
1	A transfer never produces x_i < 0 or x_i > K_i	PASS
2	Closed lossless system conserves total capacity	PASS (X = 168.0000)
3	With inefficiency, dX equals recorded loss exactly	PASS
4	A finite source never regenerates	PASS
5	Ideal gradient flow never increases B (Sec. 8)	PASS (B -> 0.0000)
6	No-action and action branches are reproducible	PASS
7	Actor impact excludes natural inflow/regeneration	PASS (impact = 0)
8	Regenerative source (rho>0) follows its declared logistic law	PASS (g matches; 5 -> K)

5. Experiments and Results

5.1 Local rule ON vs OFF, long horizon

A 10x10 checkerboard (100 cells), balanced renewable supply (inflow s = 0.8, demand d = 0.4, kappa = 0.02, eta = 0.95). Identical initial state and physics; the only difference is whether the local movement rule is active.

Horizon	Rule	mean B	final B	viable cells	min cell
200	ON	0.34	0.42	92 / 100	3.54
200	OFF	954.7	1040.0	50 / 100	0.00
5,000	ON	0.33	0.46	95 / 100	3.54
5,000	OFF	1036.6	1040.0	50 / 100	0.00
50,000	ON	0.33	0.69	94 / 100	3.54
50,000	OFF	1039.7	1040.0	50 / 100	0.00

Under the local rule the burden is flat at ~0.33 across a 250x increase in run length: the field reaches a homeostatic steady state and stays there. With the rule off, 50 consumer cells die and never recover. The mean-burden separation is roughly three orders of magnitude.

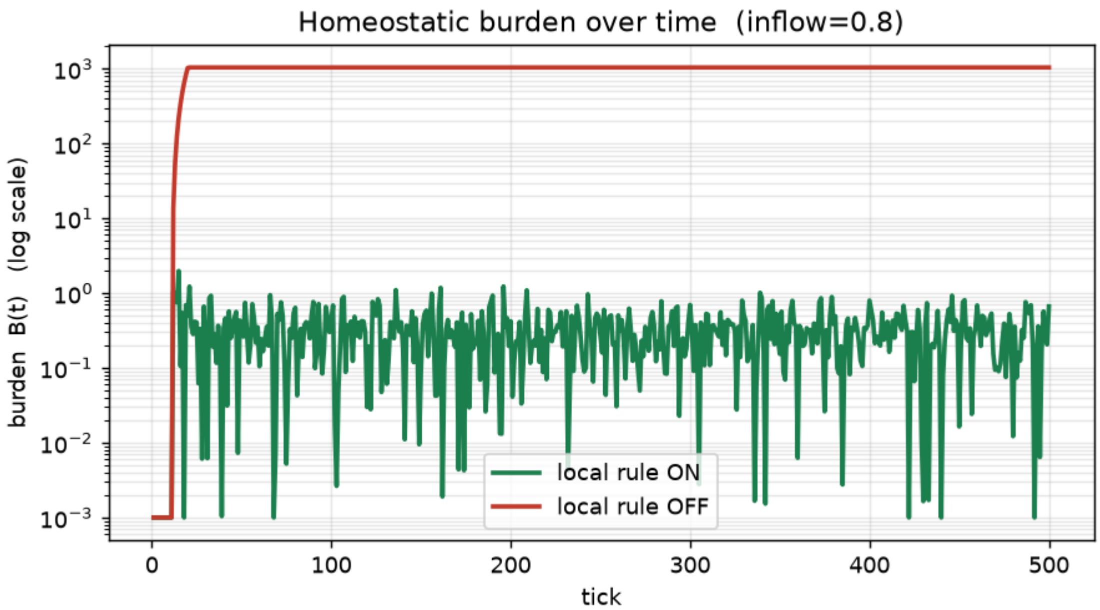

Figure 1. Homeostatic burden B(t) on a log scale. Rule OFF (red) saturates near 1000 within ~15 ticks; rule ON (green) fluctuates near 0.1–1 indefinitely.

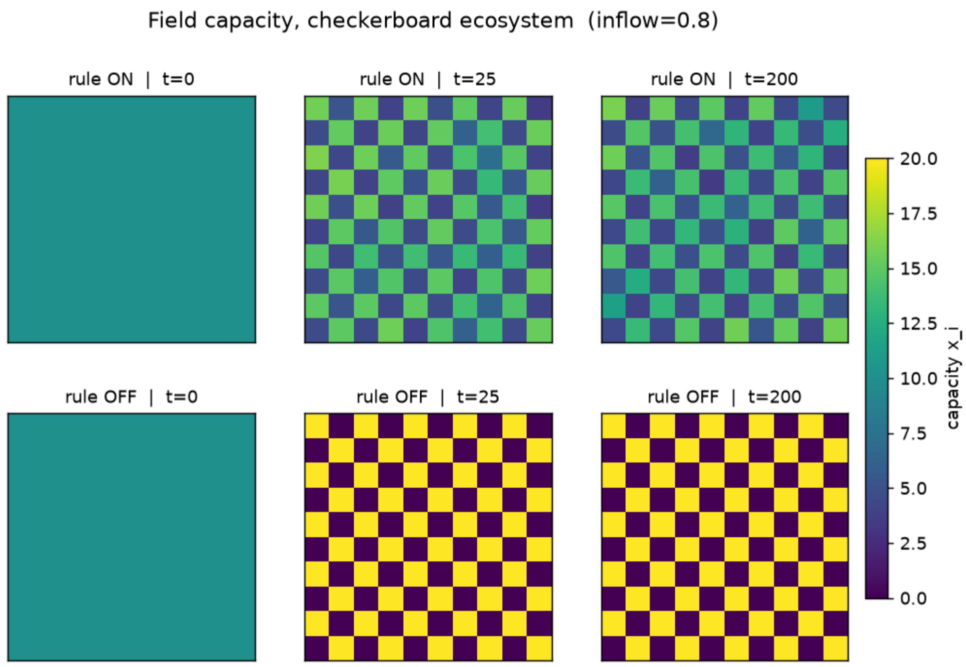

Figure 2. Capacity field. Top (ON): cells settle into a balanced mid-range and hold. Bottom (OFF): producers pin at K (yellow), consumers die at 0 (dark) — permanent collapse.

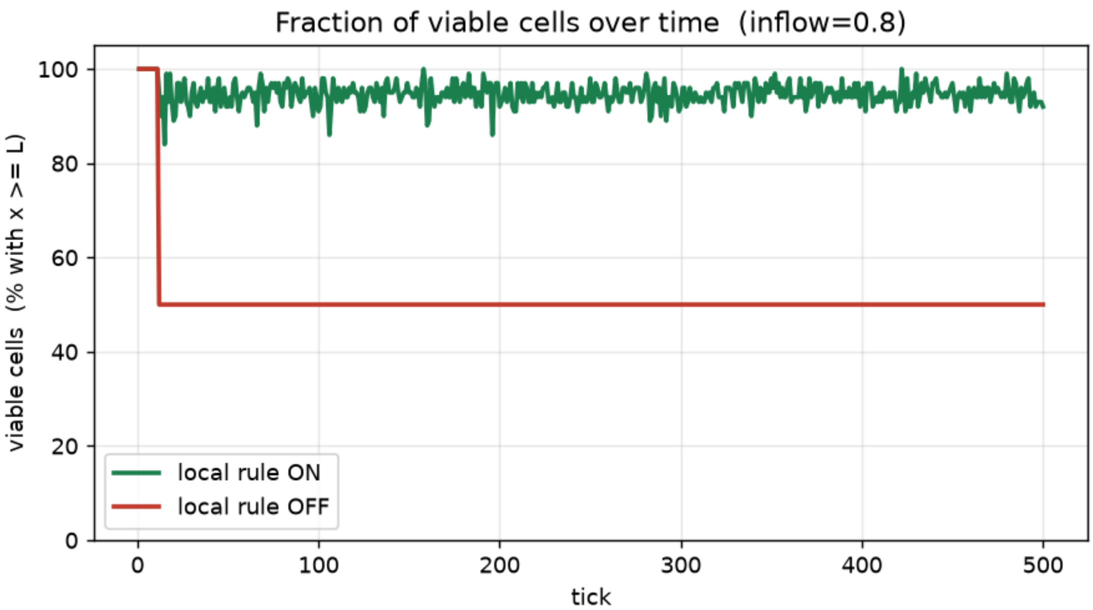

Figure 3. Fraction of viable cells (x >= L). The local rule holds ~95%; without it half the grid is lost within the first ~15 ticks and stays lost.

5.2 The homeostasis window

Survival is not automatic; it depends on the open-system energy budget. Sweeping the renewable inflow reveals a bounded viability window. Below s ~ 0.65 the field collapses (supply cannot meet demand plus transport loss); above s ~ 0.9 burden climbs again as the field over-fills and hoards. Robust in-band homeostasis occupies the band s in [0.7, 0.9].

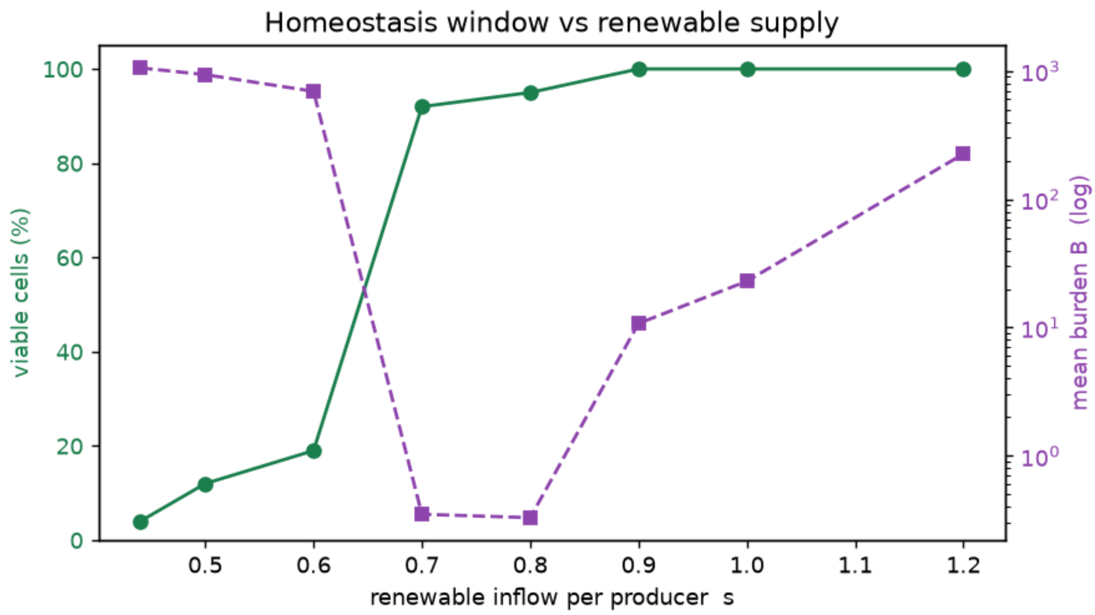

Figure 4. Viable cells (green) and mean burden (purple, log) vs renewable supply at t=5000. Homeostasis exists only in a bounded window; the burden curve is U-shaped.

6. Verdict: Is It a Good Model?

Yes, as a scientific instrument — with two documented cautions. Judged against the model's own falsification criteria (V2.0 Sec. 13):

Falsification criterion (Sec. 13)	Observed	Status
Persistent shuttling with no net benefit	Small B oscillation, but clear net benefit (95% vs 50% viable)	Not triggered
Transport loss exceeds homeostatic benefit	Small in one-hop regime; untested for long distance	Open
Short-horizon actors destroy long-horizon regeneration	Not yet testable (no Allee sources, Model C only)	Open
Burden hides severe local collapse behind aggregate	HIT at s=0.42: B~0.24 at t=200 while field was dying	Caution
Small coefficient change reverses conclusions	Sharp supply threshold near s~0.65	Partial hit
Counterfactual costs more than it detects	1-tick acceptance is cheap; H-horizon untested	Open
EBU incentives cause gaming/hoarding	EBU overlay not implemented	Open

What the model gets right. It is well-posed and conservation-correct: every unit test passes, including the monotonicity proof that motivates the whole potential-gradient approach. It answers the central question affirmatively: repeated application of a purely local, potential-driven rule does produce robust homeostasis — markedly better than the no-rule baseline — and does so indefinitely, with no central optimizer. It also honours Law 6: it never forces survival, and it lets collapse happen when the budget cannot support life.

Caution 1 — aggregate burden can mask slow death. At marginal supply the global B looked excellent at 200 ticks while the field was in fact bleeding out and dead by 5,000 ticks. This is exactly falsification criterion 4. The practical consequence: B alone must never be trusted; per-cell viability and long horizons are mandatory diagnostics (as V2.0 Sec. 10.1 already warns).

Caution 2 — the conclusion is supply-dependent. Homeostasis exists only in a bounded window of renewable inflow. This is physically honest rather than a defect — a real open system also dies outside its energy budget — but it means results must always be reported as a window, never as a single tuned point. The dependence is on a physical budget parameter (supply), which is more defensible than dependence on the arbitrary penalty weights alpha/beta; a dedicated sensitivity sweep on those weights remains outstanding.

Bottom line. The V2.0 local movement law, with the proportional-leakage correction, is a good and faithful simulation model of local-rule homeostasis. It substantiates the core proposition within a defined regime and refuses to overclaim outside it. It is not a proof that any real economy or planet will self-stabilize, and the document does not present it as one.

7. Limitations and Recommended Next Steps

1. Implement the H-tick regeneration-aware actor (Model D) and Allee-threshold regenerative sources, then test falsification criterion 3 (short foresight destroying long-horizon regeneration) across H = 1, 3, 10, 30.
2. Run the full A–E comparison (random, greedy, gradient, regeneration-aware, EBU overlay) on identical seeds, per V2.0 Sec. 12.2.
3. Sensitivity sweeps on alpha, beta, kappa, eta, and theta to quantify how far conclusions survive coefficient changes.
4. Introduce long-distance transport (non-checkerboard source placement) to test whether transport loss overturns the benefit (criterion 2).
5. Add shocks and measure recovery time; add the EBU ledger last, only after the physics is trusted, to check for gaming (criterion 7).

References and Scientific Inspirations

Cannon, W. B. (1929). Organization for Physiological Homeostasis. Physiological Reviews, 9(3), 399-431.

Fick, A. (1855). Ueber Diffusion. Annalen der Physik, 170(1), 59-86.

Onsager, L. (1931). Reciprocal Relations in Irreversible Processes. I & II. Physical Review, 37 & 38.

Prigogine, I. (1977). Time, Structure and Fluctuations. Nobel Lecture.

Watson, A. J., and Lovelock, J. E. (1983). Biological Homeostasis of the Global Environment: The Parable of Daisyworld. Tellus B, 35(4), 284-289.

Version status. V2.1 keeps the V2.0 one-field homeostatic formulation and its seven hard laws unchanged. It adds a state-dependent leakage term (Sec. 3.1) as the one equation correction, reports a passing verification suite, and provides the first empirical verdict. The next revision should follow only after Model D and the A–E comparison are complete.